

Nassau Candy Distributor

Factory-to-Customer Shipping Route Efficiency Analysis

EDA, Insights, and Recommendations

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1. Project Overview and Objective

The objective of this project is to convert raw order and shipment data into route-level logistics intelligence for Nassau Candy Distributor. The analysis identifies which factory-to-customer routes perform efficiently, which routes show higher lead time, and where route performance should be investigated further.

The dashboard and report are designed to support data-driven logistics review across factories, customer regions, states/provinces, and shipping modes. The final output helps operations teams move from reactive shipment review to structured performance monitoring.

Data note: Shipping Lead Time is calculated as Ship Date minus Order Date using the dates in the supplied dataset. The source data contains Ship Date values from 2026 to 2030 while Order Date values run from 2024 to 2025. Because actual customer delivery dates and carrier scan events are not included, this report treats the metric as recorded order-to-shipment lead time and identifies bottleneck candidates rather than proving infrastructure causes.

2. Data Cleaning and Validation

The processed dataset contains 10,194 shipment records. Order dates range from 2024-01-02 to 2025-12-31, and recorded ship dates range from 2026-06-30 to 2030-06-28. After validation, there were no missing Order Date or Ship Date values and no negative lead times.

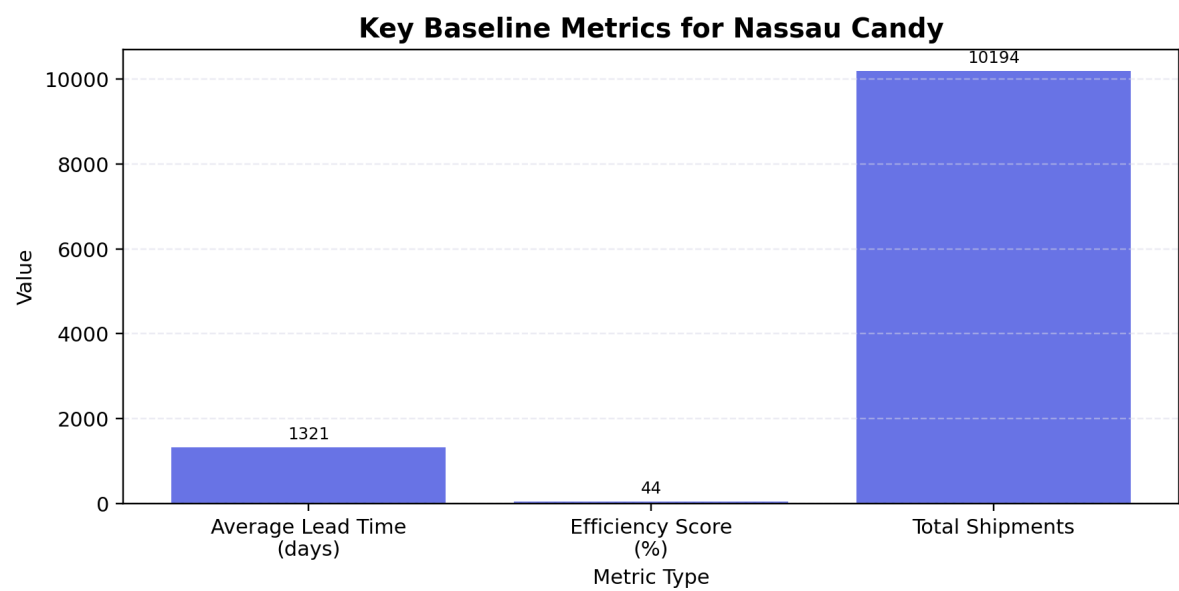
Product names were mapped to factory locations using the supplied product-to-factory correlation table. This mapping made it possible to define a route as Factory -> Customer State/Province and to compare route performance across factories, regions, and ship modes.

Check	Result
Total records analyzed	10,194
Missing Order Date / Ship Date	0 / 0
Negative lead-time records	0
Countries represented	United States, Canada
Customer states/provinces	59

3. Exploratory Data Analysis and Baseline KPIs

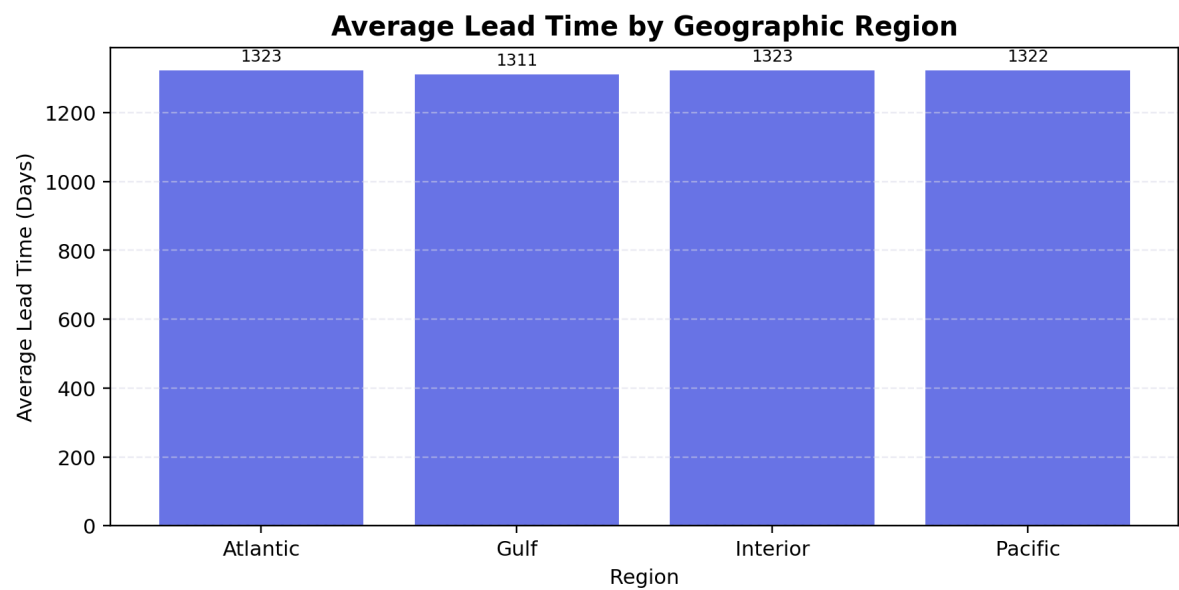
The average recorded lead time across the network is 1,320.8 days, with a median of 1,274 days. The minimum value is 904 days and the maximum value is 1,642 days.

Using min-max normalization, the overall route efficiency score is 43.5%. When the system average is used as the delay threshold, 33.1% of shipments exceed the threshold.



4. Geographic and Route Performance

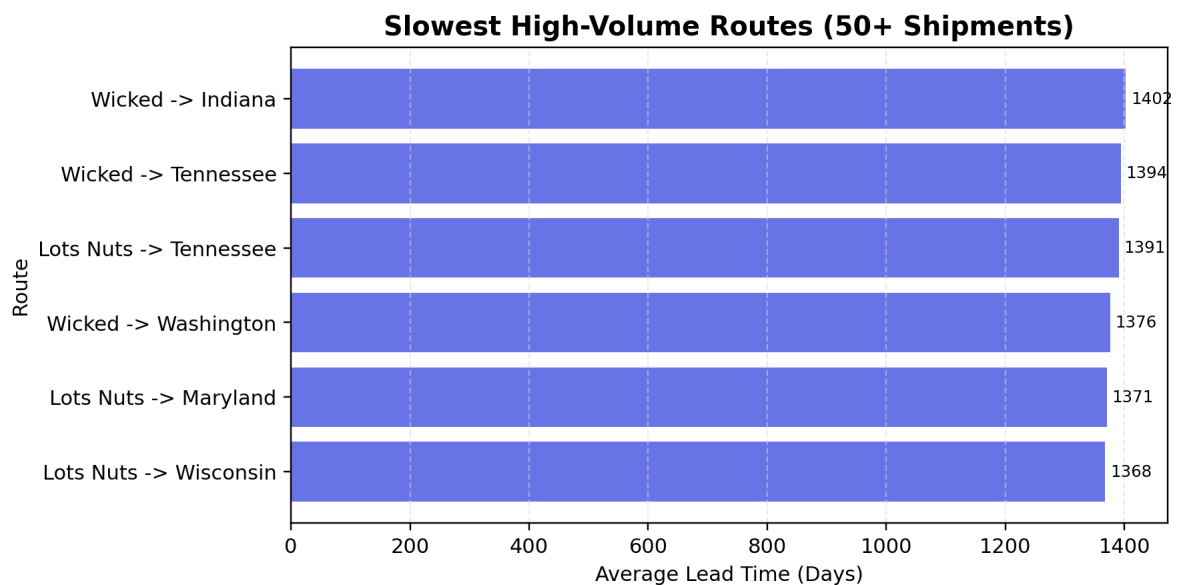
Average lead time is similar across regions, but the Gulf region performs slightly better than the other regions. The Interior, Atlantic, and Pacific regions are close to the overall system average. This suggests that route-level and factory-level analysis is more useful than relying only on broad regional averages.



4.1 High-Volume Slow Routes

To avoid overreacting to one-off shipments, the route ranking below uses routes with at least 50 shipments. These routes combine meaningful volume with higher-than-average recorded lead time and should be prioritized for operational review.

Route	Shipments	Avg Lead Time	Delta vs System Avg
Wicked Choccy's -> Indiana	54	1,402.5	81.6
Wicked Choccy's -> Tennessee	67	1,393.8	72.9
Lot's O' Nuts -> Tennessee	109	1,391.0	70.1
Wicked Choccy's -> Washington	221	1,375.7	54.9
Lot's O' Nuts -> Maryland	60	1,370.7	49.8
Lot's O' Nuts -> Wisconsin	58	1,367.6	46.8



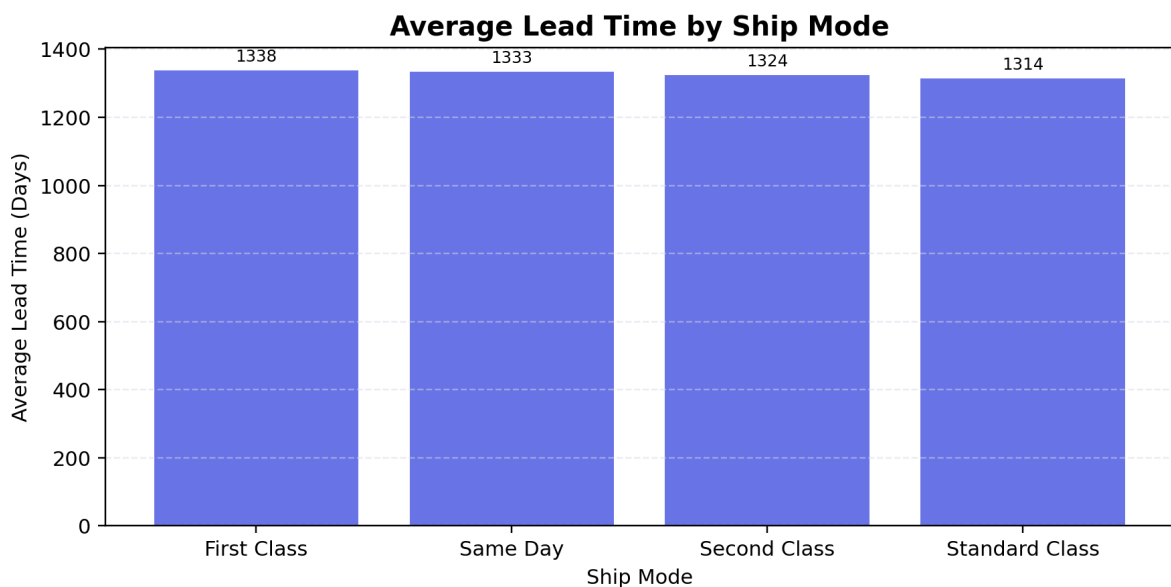
4.2 High-Volume Bottleneck Candidates

A bottleneck score was calculated as shipment volume multiplied by average lead time. This highlights routes that may have the greatest total operational impact, even when their average lead time is not the highest in the dataset.

Route	Shipments	Avg Lead Time	Bottleneck Score
Lot's O' Nuts -> California	1,125	1,316.9	1,481,524.0
Wicked Choccy's -> California	823	1,320.3	1,086,629.0
Lot's O' Nuts -> New York	645	1,330.5	858,183.0
Lot's O' Nuts -> Texas	569	1,310.5	745,652.0
Wicked Choccy's -> New York	436	1,311.9	571,981.0
Wicked Choccy's -> Texas	388	1,315.6	510,456.0

5. Ship Mode and Factory Analysis

The ship mode comparison shows that First Class and Same Day do not have the lowest recorded lead time in the dataset. This should not be interpreted as proof that the carrier service is slow. Since the data measures the time from order date to ship date, the result may reflect factory processing, fulfillment queue priority, routing allocation, or date-recording behavior before the shipment leaves the facility.



Ship Mode	Shipments	Avg Lead Time	Median Lead Time	Avg Cost	Avg Profit
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First Class	1,548	1,338.3	1,272.0	4.7	9.1
Same Day	547	1,333.4	1,269.0	4.4	8.6
Second Class	1,979	1,323.8	1,273.0	4.8	9.3
Standard Class	6,120	1,314.3	1,274.0	4.7	9.2

Factory-level results show only modest differences among major factories, with The Other Factory recording the lowest average lead time. Sugar Shack has the highest average but also very low shipment volume, so its results should be interpreted carefully.

Factory	Shipments	Avg Lead Time	Median Lead Time	Std Dev
Sugar Shack	33	1,340.0	1,274.0	265.6
Secret Factory	217	1,321.9	1,274.0	289.1
Lot's O' Nuts	5,692	1,321.2	1,274.0	261.6
Wicked Choccy's	4,152	1,321.1	1,274.0	262.2
The Other Factory	100	1,280.3	1,273.0	259.7

6. Dashboard Capabilities

The Streamlit dashboard provides an interactive interface for monitoring route performance. Users can filter by date range, factory, region, state/province, and ship mode. A lead-time threshold slider allows users to classify shipments as delayed based on the selected threshold.

- Route efficiency overview with average lead time, median lead time, shipment volume, delay percentage, and efficiency score.
- Route leaderboard showing top and bottom routes based on normalized efficiency score.
- Geographic bottleneck view using state-level heatmaps and bottleneck tables.
- Ship mode comparison using average lead time, delay rate, and distribution charts.
- Order-level drill-down with searchable shipment records and a filtered CSV download option.

7. Strategic Recommendations

- Prioritize high-volume routes first. Routes such as Lot's O' Nuts -> California and Wicked Choccy's -> California have the highest total bottleneck impact because of their shipment volume.

- Investigate high-volume slow routes. Wicked Choccy's -> Indiana, Wicked Choccy's -> Tennessee, and Lot's O' Nuts -> Tennessee show above-average recorded lead time and enough volume to justify operational review.
- Review priority shipping workflows. First Class and Same Day should be checked to ensure that priority orders receive priority treatment during factory processing and order release.
- Use the dashboard for monthly performance review. The threshold slider should be set using business rules, such as system average lead time or the 75th percentile.
- Avoid unsupported causal claims. Additional data, such as carrier transit scans, delivery dates, weather, road congestion, and warehouse queue times, would be needed before concluding that delays are caused by external infrastructure.

8. Conclusion

This project establishes a route-level view of recorded shipping lead time for Nassau Candy Distributor. The analysis shows that broad regional averages are not enough to identify operational pressure points. High-volume routes create the largest total impact, while some priority shipping modes require further investigation because they do not show lower recorded lead time in the dataset. The dashboard gives stakeholders a practical tool for monitoring performance, identifying bottleneck candidates, and improving logistics decision-making.